

Tropical Cyclone Genesis Theories

MEA 458/NCSU

By the end of the lecture, students should be able to:

- ▶ distinguish **favorable conditions** from a **genesis mechanism**;
- ▶ explain why TC genesis is a **finite-amplitude, multiscale transition**;
- ▶ Recall key aspects of different theories of TC genesis
- ▶ describe how moisture, vorticity, shear, and surface fluxes interact during genesis;
- ▶ identify the major unresolved scientific questions.

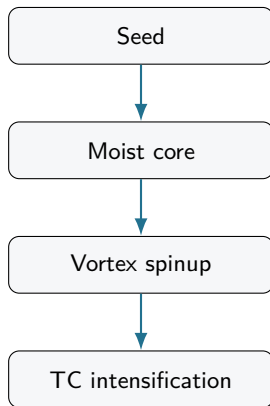
Tropical cyclone (TC) Genesis

TC genesis, as we discussed in last lecture, is still a relatively uncommon phenomenon (given favorable conditions and plenty of seed disturbances, only a fraction of them develop into a TC)

The emergence of a coherent storm system with a distinct structure (eye/eye-wall/rain-bands) points to an underlying instability (growth mechanism)

Source: Emanuel (1989, 2018); Tang et al. (2020); Montgomery and Smith (2017); Rajasree et al. (2023)

Tropical cyclone Genesis



Tropical cyclone genesis is best described as a finite-amplitude transition in which a disturbance becomes moist, acquires rotation, and sufficiently protected to develop a self-sustaining vortex with a strong, near-surface peak.

Source: Emanuel (1989, 2018); Tang et al. (2020); Montgomery and Smith (2017); Rajasree et al. (2023)

What do we mean by genesis?

Operational definitions vary by agency, but scientifically genesis means the emergence of a coherent, surface-based, warm-core cyclonic vortex.

See [this recent review of TC genesis forecast in operations](#)

Operational view

- ▶ threshold surface winds;
- ▶ closed low-level circulation;
- ▶ persistent organized convection.

Dynamical view

- ▶ vertically coupling vortex;
- ▶ protected moist core;
- ▶ sustained convective vorticity production;
- ▶ boundary-layer spinup.

Why genesis is hard

- ▶ Convection is intermittent, three-dimensional, and cloud-scale.
- ▶ The precursor disturbance is synoptic or mesoscale.
- ▶ Moisture, vorticity, surface fluxes, and radiation interact nonlinearly.
- ▶ Many favorable disturbances fail; some less obvious disturbances develop.

cloud scale $\longrightarrow$ mesoscale organization $\longrightarrow$ cyclone-scale vortex

Genesis is an upscale-organization problem plus a finite-amplitude threshold problem.

Source: Emanuel (2018); Montgomery and Smith (2017); Tang et al. (2020)

Gray's environmental framework remains the standard starting point.

- ▶ warm ocean / large air-sea disequilibrium;
- ▶ sufficient planetary vorticity;
- ▶ pre-existing low-level disturbance;
- ▶ high midtropospheric humidity;
- ▶ conditional instability;
- ▶ weak vertical wind shear;
- ▶ low environmental deformation;
- ▶ persistent convective organization.

$$Probability(\text{genesis}) = F(\eta, V_p, RH, \text{shear}, \text{ventilation}, \text{seed}, \dots)$$

Source: Gray (1968); Emanuel and Nolan (2004); Camargo et al. (2007); Tang and Emanuel (2012)

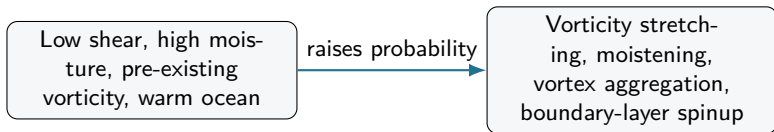
Conditions are not mechanisms

Favorable environment

Tells us where and when genesis is statistically more likely.

Genesis mechanism

Explains how a weak disturbance becomes a vertically coherent surface vortex.



A favorable environment is necessary in practice, but not sufficient.

Key concepts discussed

- ▶ The Bister and Emanuel model of TC formation from a precursor MCS
- ▶ Marsupial paradigm of TC formation
- ▶ Spontaneous aggregation of convection and TC formation

Although not necessarily mutually exclusive, the literature on TC formation often is divided into the so-called *top-down* and *bottom-up* views.

Top-down versus bottom-up genesis

Two historical descriptions are often used:

Top-down emphasis

- ▶ mesoscale convective systems;
- ▶ midlevel vortices;
- ▶ stratiform heating;
- ▶ downward extension of circulation.

Bottom-up emphasis

- ▶ low-level convergence;
- ▶ convective vorticity stretching;
- ▶ VHTs;
- ▶ surface vortex aggregation.

Better synthesis: genesis vertically couples low-, mid-, and upper-tropospheric circulation into one coherent warm-core vortex.

Source: Raymond et al. (2011); Montgomery and Smith (2017); Emanuel (2018)

The Top-Down Shower Head View

Consider a precursor disturbance such as an easterly wave that is accompanied by moist convection. The convection consists of a bunch of thunderstorms organized on the mesoscale - An Mesoscale convective system (MCS). The MCS can be thought of being composed of a few deep convective cores surrounded by a wide area of stratiform clouds and precipitation.

How does this region of convection yield a tropical cyclone?

The genesis problem: Bister and Emanuel (1997)

Question

How does a weak tropical disturbance become a self-sustaining, surface-based warm-core vortex?

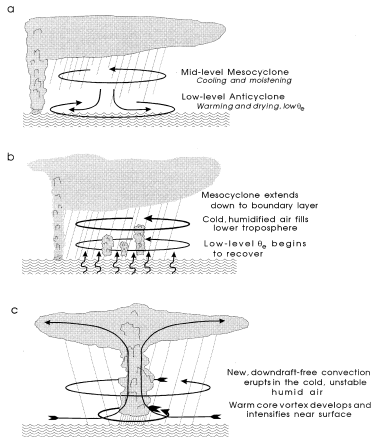


FIG. 13. Conceptual model of tropical cyclogenesis from a preexisting MCS. (a) Evaporation of stratiform precipitation cools and moistens the upper part of the lower troposphere; forced subsidence leads to warming and drying of the lower part. (b) After several hours there is a cold and relatively moist anomaly in the whole lower troposphere. (c) After some recovery of the boundary layer θ_e , convection redevelops. See text for details.

Question

How does a weak tropical disturbance become a self-sustaining, surface-based warm-core vortex?

- ▶ Mature tropical cyclones are maintained by air–sea thermodynamic disequilibrium.
- ▶ Genesis requires an incipient circulation strong enough to organize convection and surface flux feedbacks.
- ▶ Bister and Emanuel emphasized a pathway through a **cold-core midlevel vortex** before surface spinup.



- ▶ The study analyzed the genesis of eastern North Pacific Hurricane Guillermo during TEXMEX.
- ▶ Aircraft observations documented a compact surface cyclone forming beneath a preexisting cold-core cyclonic circulation aloft.
- ▶ The authors then used an idealized nonhydrostatic, axisymmetric model to test whether a midlevel rain-cooled vortex could evolve into a tropical cyclone.

Core claim

A rain-induced, cold-core midlevel vortex can moisten and precondition the column, then transition toward a warm-core, surface-intensifying cyclone.

- ▶ The ordinary tropical troposphere is not saturated through a deep layer.
- ▶ Dry midlevel air favors evaporatively cooled downdrafts, which disrupt low-level organization.
- ▶ Repeated rain and convection moisten the lower and middle troposphere.
- ▶ Once downdrafts weaken, convection couples more effectively to the low-level circulation.

Deep Moist Column

Genesis is favored as the column approaches a deep, nearly saturated, moist-adiabatic state.

Simple interpretation

The disturbance first has to “remove the dry-air penalty” before surface-flux feedbacks can efficiently amplify the vortex.

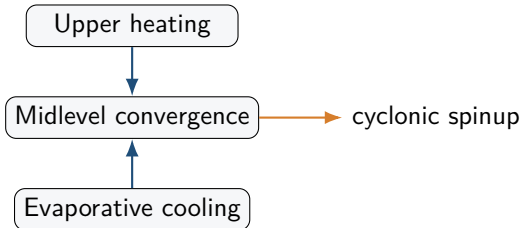
How the midlevel vortex forms

- 1 Stratiform precipitation falls into relatively unsaturated air.
- 2 Evaporation and melting cool the lower-to-middle troposphere and increase humidity.
- 3 Diabatic heating is vertically nonuniform: heating above, cooling below.
- 4 The balanced response produces convergence and vortex stretching near the midlevels.

Vorticity stretching

$$\frac{D\zeta_a}{Dt} \approx -\zeta_a \nabla_h \cdot \mathbf{v}$$

where $\zeta_a = f + \zeta$. Convergence increases cyclonic absolute vorticity.



Why the vortex is initially cold-core and elevated

- ▶ Evaporative cooling lowers temperature in the precipitating column. But this is not the only reason.
- ▶ With a cyclonic vortex peaking in the mid-level, thermal wind balance demands that the vortex is cold core below the vortex and warm core above. Recall our discussion of cold and warm core disturbances from Atmospheric Dynamics.
- ▶ The resulting vortex is therefore not initially a mature tropical cyclone: it is an elevated, cold-core mesoscale cyclone.

Key distinction

This is not the classical picture of immediate low-level warm-core amplification. It is a two-stage route: first a midlevel cyclonic reservoir, then surface-based intensification.

How the vortex “descends” to the surface

Key Point

The vortex does not simply sink as a rigid object. The circulation develops downward as the thermodynamic and dynamical coupling shifts toward the boundary layer.

- 1 Column moistening suppresses evaporatively driven downdrafts.
- 2 Convection becomes deeper and more persistent near the vortex center.
- 3 The level of strongest convective heating and convergence lowers as the lower troposphere saturates.
- 4 Boundary-layer inflow strengthens beneath the midlevel vortex.
- 5 Surface enthalpy fluxes increase with wind speed, initiating WISHE-type amplification (see next slide).

Surface Enthalpy Flux: The Emanuel View

Ocean as an energy source

In Emanuel's framework, the tropical cyclone is a **heat engine**: the warm ocean supplies energy to the atmospheric boundary layer through **surface enthalpy flux**.

The turbulent enthalpy flux from the ocean is approximately

$$F_k = \rho C_k |\mathbf{V}| (k_s^* - k_a)$$

- ▶ k_s^* : saturation enthalpy of air at the sea-surface temperature
- ▶ k_a : enthalpy of boundary-layer air
- ▶ $|\mathbf{V}|$: surface wind speed
- ▶ C_k : surface exchange coefficient for enthalpy

Stronger winds $\Rightarrow$ larger enthalpy flux $\Rightarrow$ warmer/moister boundary layer $\Rightarrow$ stronger convection $\Rightarrow$ lower pressure + stronger winds

Key idea: WISHE

This creates a positive feedback: **Wind-Induced Surface Heat Exchange (WISHE)**. Once a sufficiently organized vortex exists, increasing surface winds allow the storm to extract progressively more energy from the ocean.

Emanuel (1986, 1995, 2003); Rotunno & Emanuel (1987)

Surface enthalpy flux

$$F_k \approx \rho C_k |V| (k_s^* - k_a)$$

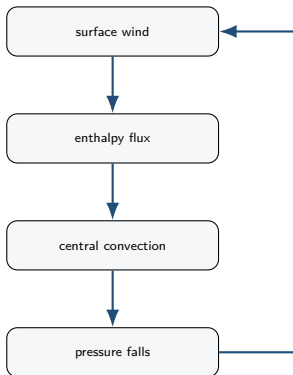
Stronger surface winds increase ocean-to-atmosphere enthalpy flux.

- ▶ Larger fluxes support stronger convection.
- ▶ Stronger convection lowers surface pressure and enhances inflow.
- ▶ Stronger inflow increases surface winds.

Surface-flux feedback after preconditioning

Key idea: WISHE

This creates a positive feedback: **Wind-Induced Surface Heat Exchange (WISHE)**. Once a sufficiently organized vortex exists, increasing surface winds allow the storm to extract progressively more energy from the ocean.



Conceptual vertical evolution

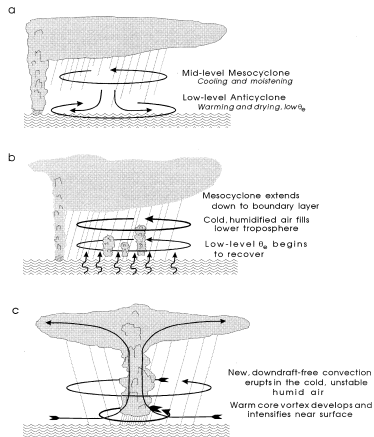


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What the theory explains well

- ▶ Why genesis often follows periods of persistent, organized convection rather than isolated deep convective bursts.
- ▶ Why midlevel humidity is critical: dry air promotes downdrafts that oppose low-level spinup.
- ▶ Why stratiform precipitation can be dynamically constructive even when it contains evaporative cooling.
- ▶ Why the transition to a tropical cyclone is abrupt once the surface circulation can access air–sea enthalpy flux feedbacks.

- ▶ Later work retained the importance of a moist protected region but often emphasized rotating wave “pouches,” vortical hot towers, and convective-scale vorticity aggregation.
- ▶ Bister–Emanuel is most naturally viewed as a **thermodynamic–balanced-dynamics pathway**: stratiform rain moistens and cools the column, creating a midlevel vortex that enables surface-flux feedback.
- ▶ Modern interpretations differ on whether the midlevel vortex is essential or one route among several.

- 1 Stratiform rain and evaporative cooling can create an elevated cold-core mesoscale vortex.
- 2 The vortex and precipitation moisten the column, reducing destructive downdrafts.
- 3 Convection becomes more vertically coupled to the boundary layer.
- 4 Low-level convergence and surface winds strengthen beneath the midlevel circulation.
- 5 Surface enthalpy flux feedback converts the preconditioned disturbance into a warm-core tropical cyclone.